TradeHub v2.1 — MVP & Implementation Plan

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Summary

TradeHub v2.1 ships an Equities-only MVP in about 16 part-time weeks: news pipeline, Dashboard, Ticker Pages and a daily-bar backtester, with Hermes evals as part of the pass/fail gate. Paper sim, morning briefing, Deep Analysis and the suggestion loop follow in v1.0.5 and v1.0.8; Crypto, Commodities and Smart Money come after.

Status: scope locked, ready for Phase 0. Replaces v2.0 (2026-11-10).

Owner: single user, Portugal (EU). Self-hosts Docker + n8n. Solo, part-time. Budget €0–5/month. No broker, no live money.

What changed from v2.0

- Smaller MVP: Equities pipeline + Dashboard + Ticker Pages + daily backtester. Paper sim, briefing, Deep Analysis and suggestions moved later.
- Timeline about 16 weeks for the MVP instead of 8.
- New Phase 0.5: every data source gets a working key and one successful call before UI work.
- News and price recording starts in week 3, to build history early.
- Price history stored as frozen, versioned Parquet snapshots in DuckDB. MVP backtests use ETFs only; futures follow as separate configs with a defined roll method. PostgreSQL, self-hosted in Docker, holds app data only (replaces Supabase).
- Historical data plan and 6 starter strategies added, including two price-action strategies.
- Evaluation setup added, same structure as Receipt Ghostbuster.
- Source changes: Tiingo, BLS, Binance (v1.1) and a one-time Databento backfill added; StockTitan, Brave, Alpha Vantage, Massive and the Treasury API not used; Reddit and Polymarket optional, with news-only sentiment and a FRED-only Fed panel as defaults.

Locked decisions

Decision	Choice
MVP scope	Equities: pipeline, Dashboard, Ticker Pages, daily backtester, Hermes evals

Decision	Choice
Stack	React + Vite, FastAPI, PostgreSQL in Docker, DuckDB + Parquet, n8n, Docker
Timeline	MVP gate at week 16
Starter strategies	Gap fade, price action supply/demand, VIX spike reversion, FOMC drift, gold + dollar + COT; SLC as alternative
Orchestration	n8n only triggers jobs; all logic in Python
Time	Everything stored in UTC; market sessions from a calendar library

What TradeHub is

A personal web app that collects free market news and data, ranks what matters each day, has Hermes explain it in plain words, and tests the owner's strategies on past data. The owner makes every trade by hand elsewhere.

It covers three categories, added one at a time: Equities (MVP), Crypto, Commodities. Smart Money (insider filings; Polymarket whales only if reachable) is an on-demand section.

Core principles (non-negotiable)

1. **Code computes, Hermes concludes.** Scores, rankings, indicators and backtests are deterministic code. Hermes only condenses, narrates and suggests. Hermes never invents numbers, and an automatic check proves it.
2. **Strategies are data.** Every strategy is a JSON config. Changing a parameter needs no code.
3. **Tags from a fixed menu.** Drivers, scheduled flag and asset map are config tables, so they are queryable and backtestable.
4. **Fail honest.** Source down = real error + alert + stale label. Never fake data; demo data only with a visible label.
5. **Measure before trusting.** Hermes outputs, the tagger and every signal are evaluated against labeled data before they drive anything.
6. **Solo-dev discipline.** Small vertical slices; each version ships with tests and a changelog. One finished category beats three half-built.
7. **€0–5/month.** Every source must justify its maintenance cost.

The 7 sections and the MVP cut

The MVP builds five of the seven sections, Equities only; the rest arrive in later versions.

#	Section	Contents	Version
1	Dashboard	Futures & macro strip (ES, NQ, YM, RTY, VIX, DXY, US10Y, GC, CL, SI), Fed & macro events panel with countdowns, hot news ranked, event calendar incl. earnings, market mood, watchlist	v1.0 (favorite strategies + briefing in v1.0.5)
2	Market Pulse	Tabs Equities / Crypto / Commodities; per-source sentiment; sector heatmap from sector ETFs	v1.0 Equities tab; v1.1 Crypto; v1.2 Commodities
3	Ticker Page	Any symbol: news, sentiment, driver tags, levels, upcoming earnings, backtest and paper-trade history	v1.0 (paper history in v1.0.5)
4	Strategy Lab	Strategy configs, backtests, results per strategy	v1.0 backtests; v1.0.5 paper sim + journal; v1.0.8 suggestions + A/B
5	Deep Analysis	Fundamentals per ticker: computed scores in Python, report by Hermes	v1.0.8
6	Smart Money	SEC Form 4 / 13F insider filings; Polymarket whale wallets only if reachable	v1.3
7	Settings / Sources	API keys, source toggles, scrape depth, LLM budget, source health (last success / last error)	v1.0

In the MVP (v1.0)

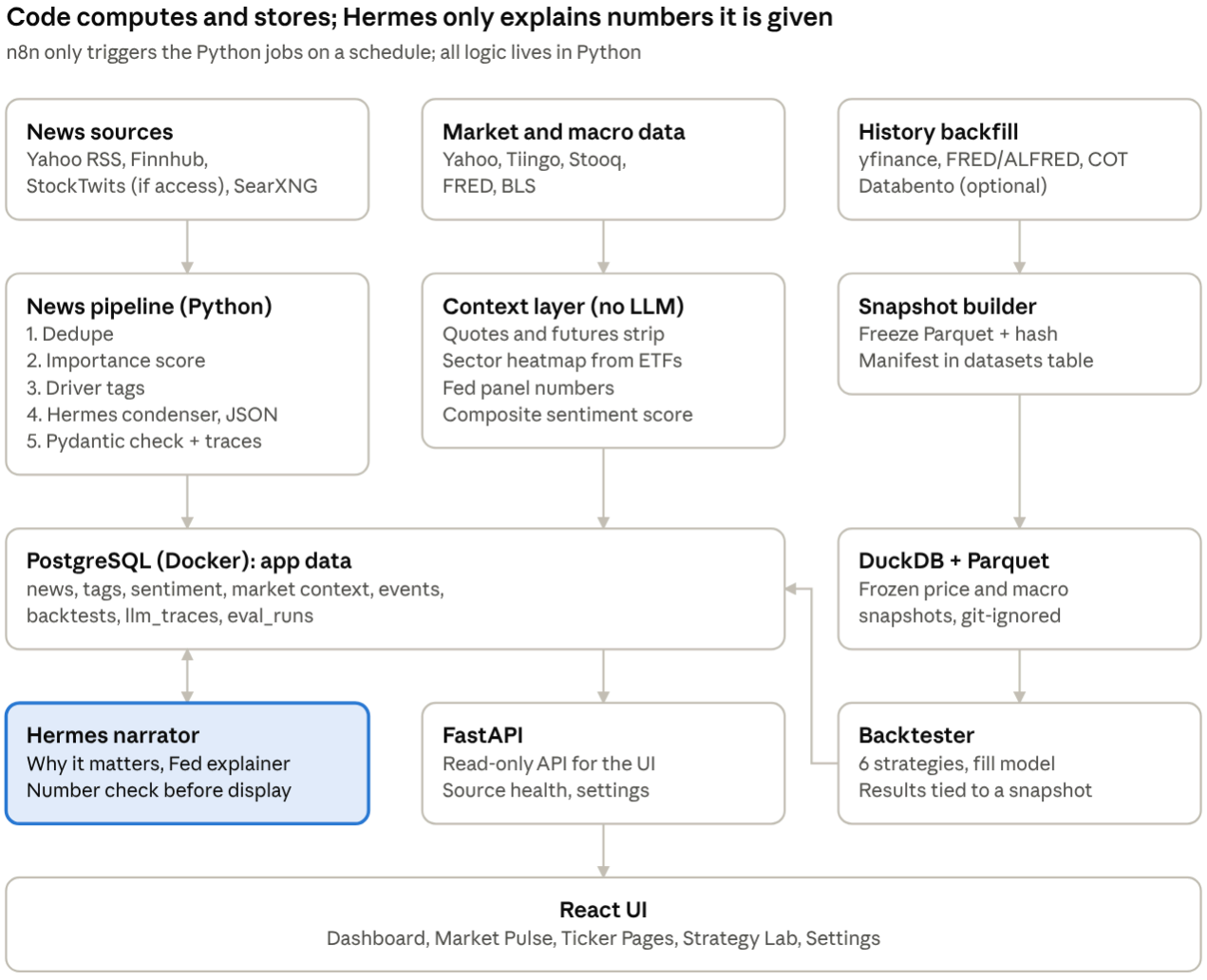
- Equities ingestion, dedupe, importance ranking, driver tags, Hermes condenser
- Hermes "why it matters" lines for the top 5 news items and Fed-panel explanations
- Dashboard (strip, Fed panel, hot news, calendar, mood, watchlist), Market Pulse Equities tab, Ticker Pages, Settings with source health

- Historical data store (DuckDB + Parquet snapshots) and a daily-bar backtester with the 6 starter strategies (SLC as the alternative)
- Full evaluation setup: code tests, Hermes evals, signal checks, CI

Not in the MVP: live paper sim, journal, morning briefing, Ask-Hermes, Deep Analysis, strategy suggester, weekly review, Crypto, Commodities, Smart Money.

Architecture and data flow

One pipeline serves every category: sources feed Python jobs, results land in PostgreSQL or frozen Parquet snapshots, and the UI only reads.



TradeHub architecture · sources to UI

Hermes touches the data twice: the condenser turns each news item into checked JSON, and the narrator explains numbers that code already computed. Everything Hermes writes passes the number check before the UI shows it.

Daily composite sentiment score (code, per ticker, per day, -100 to +100): news sentiment average + news volume + driver-tag directions from the sensitivity matrix, plus StockTwits bull/bear ratio when available. Shown as the watchlist badge and on Ticker Pages.

Schedules: equities morning-heavy crons plus a light rolling refresh; crypto every 30–60 min from v1.1; COT weekly on Friday from v1.2. Scrape depth is configurable per source, with a global "go deeper now" button; Firecrawl only as a last resort.

Tools and data sources

Every need has a main source and, where one exists, a backup; sources are config rows, so swapping one is a config change.

Data sources

Need	Main	Backup	Version	Free-tier notes
Daily prices (stocks/ETFs)	Yahoo (yfinance)	Tiingo	v1.0	Tiingo: 50 req/hour, 1,000/day, ~500 symbols/month, personal use
Futures, VIX, DXY (strip)	Yahoo	Stooq, then Twelve Data	v1.0	Stooq needs a free API key since 2026; Twelve Data 800/day
Quotes + company news	Finnhub	Yahoo RSS	v1.0	Finnhub: 60 calls/min, 1 year company news
Earnings calendar	FMP	Finnhub	v1.0	Finnhub free window verified in Phase 0.5; yfinance earnings dates as a third fallback
Fundamentals	FMP (`/stable/` endpoints)	–	v1.0.8	250 calls/day; check each endpoint on a free key
Sector heatmap	Own calculation from 11 sector ETFs (XLK, XLF, XLE…)	–	v1.0	Uses daily prices above

Need	Main	Backup	Version	Free-tier notes
Social sentiment	News-only: Hermes sentiment + news volume + tags	StockTwits if access works; Reddit only if approved	v1.0	Social sources are additive, never required
Deep search	SearXNG (self-hosted)	none	v1.0	Tavily later only if needed
Macro + rates	FRED (incl. ALFRED vintages)	BLS for release dates	v1.0	Free
Fed panel	FRED (rates, CPI, breakevens)	Polymarket odds, only if reachable later	v1.0	Polymarket is blocked in Portugal; CME FedWatch kept as an external link
Intraday history (one-time)	Databento	–	Phase 3, optional	$125 free credits for new users
Crypto prices + dominance	CoinGecko	–	v1.1	
Crypto funding rates	Binance public API	–	v1.1	
Crypto mood	Fear & Greed (alternative.me)	–	v1.1	
Positioning	CFTC COT (Socrata)	–	v1.0 history, v1.2 live	History back to 1986
Oil data	EIA	–	v1.2	
Whale wallets	Polymarket	Etherscan multichain (Polygon)	v1.3	Only if reachable from Portugal; EDGAR insider filings come first
Insider filings	SEC EDGAR	–	v1.3	
On-chain extras	DefiLlama, Bitquery	–	v1.3, optional	

Infrastructure and AI

Role	Tool
Run everything	Docker Compose
Schedule jobs	n8n (triggers only)
Backend logic + API	Python, FastAPI
Frontend	React + Vite
App database	PostgreSQL 16 in Docker (persistent volume; not exposed to the internet, FastAPI is the only client)
Price history	DuckDB + Parquet
Data work	Polars / pandas
Trading calendar	pandas_market_calendars
Cheap LLM tier	Ollama (local) or a small OpenRouter model, chosen by the model-comparison eval
Better LLM tier	OpenRouter (Hermes)
Output validation	Pydantic
Backtest cross-check	backtesting.py
Tests + CI	pytest, ruff, GitHub Actions
Errors (optional)	Sentry free tier with a privacy scrubber

Not used: StockTitan (no public API), Brave Search (no free tier since Feb 2026), Alpha Vantage (25 requests/day), Massive (overlaps Tiingo), US Treasury API (FRED covers it), Finviz, X/Twitter, CryptoPanic, IEX Cloud.

Source notes: Tiingo limits · Stooq API key · FMP v3 deprecation · Finnhub free plan · Polymarket blocked in Portugal · Reddit Responsible Builder Policy · Brave API pricing · StockTitan FAQ · Databento

Driver taxonomy and sensitivity matrix

The taxonomy stays locked at 7 tags plus a scheduled flag; the cap is about 10–12.

Tag	Meaning	Example
Geopolitical	Wars, escalations, sanctions	"Escalation in Middle East"

Tag	Meaning	Example
Monetary	FOMC, rate expectations, CPI prints, real yields	"Fed signals slower cuts"
Macro risk	Bank failures, collapse fears, debt-ceiling drama	"Regional bank under pressure"
Supply/flows	OPEC, EIA inventories, central-bank gold buying, mine outages	"Crude inventories draw"
Positioning	COT extremes, ETF flows, retail sentiment extremes	"Gold COT record long"
Dollar	DXY direction, de-dollarization narratives	"DXY +0.5% on the day"
Liquidity flush	Assets sold together in margin-call cascades	March 2020 style
scheduled flag	true = known in advance (FOMC, CPI, OPEC, COT Friday); false = surprise	On every tagged item

Driver direction: every tag also carries a direction from a fixed menu: +1, −1, or 0 (unclear, left out of the matrix maths).

Tag	+1 means	−1 means
Geopolitical	Escalation	De-escalation
Monetary	Hawkish	Dovish
Macro risk	Stress rising	Stress easing
Supply/flows	Supply tightening or inflows	Supply loosening or outflows
Positioning	Crowded long	Crowded short
Dollar	Dollar up	Dollar down
Liquidity flush	Flush under way	Not used (+1 or 0 only)

The code tagger sets the direction when keywords are clear ("rate hike" → Monetary +1). Otherwise the condenser fills `driver_direction` from the same menu, and Pydantic rejects any other value.

Sensitivity matrix: a config table mapping each driver to an expected direction and horizon per asset (war: gold up over days; liquidity flush: gold down short-term; BTC ambiguous). Every entry is a hypothesis until a signal check supports it, and is labeled that way in the UI.

Tag rule: a tag stays only if it maps to a direction/horizon and at least one source can fill it. Tagger quality is measured with precision and recall per tag (see Evaluation).

Catalyst types (fixed menu for the condenser): earnings, guidance, analyst_rating, m_and_a, product, legal_regulatory, management, macro, other. Hermes can only pick from this list; anything else fails validation and is logged.

Ticker IDs: the `tickers` table uses a canonical ID with an asset-class prefix, e.g. `FUT:GC`, `ETF:GLD`, `SPOT:XAUUSD`, `IDX:SPX`, `EQ:TSLA`. Display names like "Gold" map to one canonical ID, so names like GOLD, SIL or OIL never collide with real tickers.

Hermes layer

Hermes uses two tiers and eight templates; the MVP ships three of them, and every call is validated, logged and evaluated.

Tiers

- **Cheap tier:** high-volume news condensing to strict JSON. Ollama (local) or a small OpenRouter model, decided by the model-comparison eval in Phase 4.
- **Better tier:** OpenRouter Hermes for narration, briefings, analysis and suggestions.

Templates

#	Template	Output	Tier	Version
1	Sentiment condenser	`{catalyst_type, sentiment_score, tickers[], driver_direction, key_facts[], one_sentence_reason}` per news item	Cheap	v1.0
2	Importance narrator	"Why it matters" for the top 5 items, using only given facts	Better	v1.0
3	Fed-panel explainer	Short explanation of upcoming macro events and the latest prints	Better	v1.0
4	Morning briefing	Futures, driver mix, sector leaders, what to watch, from computed numbers	Better	v1.0.5
5	Ask-Hermes	Read-only chat over DB state ("why was AMD skipped today?")	Better	v1.0.5
6	Ticker summary	One-screen story for a Ticker Page	Better	v1.0.5

#	Template	Output	Tier	Version
7	Deep analysis report	Fundamentals narrative and risks from computed scores	Better	v1.0.8
8	Strategy suggester + hypothesis proposer	Config tweak → hypothesis row → backtest job; never auto-applied	Better	v1.0.8

Hard rules

- Strict JSON + Pydantic validation on every call; invalid → retry once → fail loud + log. Menu fields (catalyst type, driver direction) accept only values from their fixed lists, and sentiment score must be between −1 and +1.
- Hermes receives numbers already computed by code and never computes or predicts from them.
- **Number check:** every figure written in digits must match a number in the input Hermes was given; a mismatch is logged as a failure and the text is not shown. Match = a difference of at most half a step of the last decimal shown ("2.3%" matches 2.31 or 2.34, not 2.36; "2%" matches 1.5 to 2.49). Allowed: small counts up to 10, ordinals, dates and years present in the input, a sign dropped only when the wording carries it ("fell 1.2%" for −1.2), and units like 1.2B. Everything else fails. Expect about a week of tuning in Phase 2, tracked by eval_numbers.py.
- Every call logged to `llm_traces`: template, prompt version, model, latency, tokens, cost, outcome.
- Prompts are versioned files in the repo, so eval results tie to a prompt version.
- Daily cap on condenser calls; budget alert at €5/month.
- Suggestions become hypothesis rows that must pass a backtest before the owner approves them.

Database and storage

PostgreSQL in Docker holds app data (about 20 tables, reachable only inside the Docker network); DuckDB + Parquet holds price and macro history as frozen snapshots.

PostgreSQL tables

Table	Purpose	Version
`sources`	Source registry: type, credentials ref, toggle, schedule, depth, last success, last error	v1.0

Table	Purpose	Version
`news_items`	Raw items: title, url, source, `published_at`, `ingested_at`, dedupe hash, raw text ref	v1.0
`news_sentiment`	Condenser JSON per item per ticker + prompt version	v1.0
`driver_taxonomy`	The 7 tags (config)	v1.0
`driver_sensitivities`	Direction + horizon per asset (config)	v1.0
`tagged_events`	Tagged items + direction (−1/0/+1) + scheduled flag + affected assets	v1.0
`tickers`	Symbol master with canonical IDs	v1.0
`market_context`	Snapshots: futures, VIX, DXY, US10Y, sector scores, Fed rates; regime label from v1.0.8	v1.0
`economic_events`	FOMC/CPI/NFP calendar + per-ticker earnings dates	v1.0
`daily_scores`	Composite sentiment score per ticker per day	v1.0
`watchlist`	Pinned symbols	v1.0
`strategy_configs`	Strategies as data	v1.0
`datasets`	Snapshot registry: version, path, hash, date range, symbols	v1.0
`backtests`	Config hash, dataset version, instrument_type, fill params, metrics with bootstrap range, confidence label, verdict	v1.0
`llm_traces`	Every Hermes call	v1.0
`eval_runs`	Eval name, dataset, prompt + model version, scores	v1.0
`starred_strategies`	Favorites for the Dashboard	v1.0.5
`paper_trades`	Simulated entries/exits with fill details	v1.0.5
`journal`	Per-trade rows + notes	v1.0.5

Table	Purpose	Version
`hypotheses`	Suggested change → backtest → approve/reject, kept forever	v1.0.8
`pm_wallets`, `pm_trades`, `insider_filings`	Smart Money	v1.3

DuckDB + Parquet (folder `data/`, git-ignored)

- `bars_daily`, `bars_intraday`, `macro_series`, `cot` stored under dated snapshot folders, e.g. `data/v2026-11-20/`.
- Each snapshot has a manifest (sources, date range, row counts, hash) mirrored in the `datasets` table.
- Snapshots are never edited. New data = new snapshot; backtests always name the snapshot they used.

Storage hygiene: keep only headlines, links and short excerpts in PostgreSQL; prune raw article text after 30 days to keep the database small and backups fast. Backups: nightly pg_dump plus a weekly copy of Parquet snapshots to a second disk or cloud folder, with a monthly restore test.

Strategy engine and backtesting

The MVP backtester runs six starter strategies on frozen daily snapshots and must give byte-identical results on rerun; gap fade is built first to prove the engine, and live paper sim follows in v1.0.5 with the same fill rules.

Strategy-as-config contract: `universe → structure_tf + entry_tf → entry_conditions → exit_logic → risk → session`. Timeframes are config values, so the same strategy runs on weekly + daily now and on 4h + 5m once intraday data exists, with no code change. The data layer resamples base bars into each timeframe, and the engine aligns them so the higher timeframe only sees closed candles. A missing indicator becomes a small, tested Python function.

Building blocks v1 (only what the starter strategies need): swing detector (N bars each side), market-structure tracker (validated highs/lows, reversals), supply/demand zone detector, median candle range, Stochastic (14, 3, 3; levels 80/20), EMA 20/50/200, RSI, ATR, gap %, ADV, prior-day levels, VIX level/change, dollar direction (FRED broad dollar index, DTWEXBGS), COT percentile, scheduled-event flags (FOMC day, day before FOMC). Driver-tag counts become conditions once enough tag history exists.

Starter strategies

#	Strategy	Idea	Data needed	When
1	Gap fade	Large opening gap → bet on a partial fill back	Daily OHLC	Phase 3, built first to prove the engine
2	Price action supply/demand	Trade only with validated structure; enter on a pullback into a demand zone (long) or supply zone (short); minimum 2.5R	Daily OHLC	Phase 3, main strategy
3	VIX spike reversion	VIX jumps above a threshold → buy SPY, exit after N days	Daily SPY + VIX	Phase 3
4	FOMC drift	SPY behaviour on and before FOMC days	Daily SPY + backfilled FOMC dates	Phase 3
5	Gold + Dollar + COT	Gold signal filtered by dollar direction (DTWEXBGS) and extreme COT positioning	Daily GLD, FRED broad dollar index (DTWEXBGS), COT history	Phase 3
6	SLC (Structure, Level, Confirmation)	Higher-timeframe trend + supply/demand zone + Stochastic confirmation, fixed 2R target	Weekly + daily now; 4h + 5m later	Alternative: weekly/daily test in Phase 3, intraday test once intraday data exists

Price action supply/demand — exact rules (defaults, all config parameters)

Rule	Definition for code	Default
Swing high/low	Bar whose high (low) is the highest (lowest) of N bars on each side	N = 2
Uptrend / downtrend	Higher swing highs and higher swing lows / lower swing highs and lower swing lows	–

Rule	Definition for code	Default
Validated low (high)	A swing low is validated once price closes above the previous swing high (mirror for highs)	–
Reversal	Close below the validated low (uptrend) or above the validated high (downtrend)	–
Impulsive move	Candle range ≥ K × median range of the last 20 candles, closing in the trend direction	K = 2
Zone	High to low of the last candle before the impulse	1 candle
Zone use	First touch only; open zones cancelled when the trend flips	–
Entry	Limit at the zone edge, filled when price touches it	–
Stop	Beyond the far side of the zone	0.1 × median range
Target	The swing high (long) or swing low (short) made after the impulse	–
R:R filter	Skip if (target − entry) / (entry − stop) < 2.5	2.5
Same-bar stop and target	Assume the stop hit first	–
Positions	One per instrument; longs and shorts allowed in simulation	–
Markets + timeframe	MVP: SPY, QQQ, GLD (ETF proxy), daily bars · futures versions: later configs (ES, NQ, GC) · CL in v1.2	–

SLC — exact rules (defaults, all config parameters)

Rule	Definition for code	Default
Structure	Trend from swings on `structure_tf`, using only candles already closed at signal time	Weekly (later 4h)

Rule	Definition for code	Default
Consolidation = no trade	Neither a higher-high/higher-low nor a lower-high/lower-low sequence in the last 4 swings	–
Level	Supply/demand zone on `entry_tf`: last candle before an impulsive move, same detector as strategy 2	Daily (later 5m)
Valid levels	Fresh (untested) zones only; break-and-retest zones added later as a variant	–
Invalidation	Zone discarded after 2 closes through it	2
Confirmation	Price inside the zone, Stochastic %K crosses beyond 80 (20) and back below 80 (above 20)	14, 3, 3; 80/20; %K = blue line
Entry	Next candle's open after confirmation	–
Stop	Just beyond the zone	0.1 × median range
Target	Fixed 2R	2R
Day-trading mode (intraday only)	Close all positions at session end	–
Markets	MVP: SPY, QQQ (ETF proxy) · futures versions: later configs (ES, NQ)	–

Fill model: market entries fill at the next bar's open; limit entries (zone strategies) fill at the limit price when the bar touches it. Spread, slippage buffer and commissions are always applied, and when stop and target fall in the same bar the stop counts first. Parameters are reported with every result. No mid-bar guessing.

Backtester rules

- Plain deterministic loop (Polars/pandas). Same config + same snapshot = byte-identical metrics, enforced in CI.
- Each result stores: config hash, dataset version, fill parameters, code version.
- Cross-check: at least one strategy also run in backtesting.py; results must match within a small stated tolerance.

- Pass/fail uses expectancy and profit factor after costs, minimum 30 trades. Results under 100 trades are labeled "low confidence", and expectancy is reported with a bootstrap 90% range. Win rate is shown, never used to decide. Promotion thresholds are set in `docs/BACKTESTING.md` before the first run and copied from the v1.4 spec.
- Data split: tune on 2010–2020, validate on 2021–2023, final check on 2024–2026 (held out until the end). Walk-forward for parameter tuning.
- Every run is kept, including failures. A result that only works with zero costs is discarded.
- Known biases written next to results: survivorship (today's universe), and short tag history.

Optional extra metrics (Phase 3, not required for the MVP gate): Sortino, Calmar and maximum drawdown next to expectancy and profit factor. Idea borrowed from MarketRegimeTrader (MIT).

Instruments (MVP = ETFs only): MVP backtests use SPY, QQQ and GLD, so no contract rolls are needed; futures stay on the Dashboard strip as context. Every backtest result carries `instrument_type` (ETF or futures). Futures versions of a strategy are separate configs with their own costs and trading hours, and are never compared one-to-one with ETF results.

No oil in the MVP: USO is not used as an oil proxy (contango decay, 2020 change in how it holds contracts). Oil backtests wait for back-adjusted CL futures in v1.2.

Futures roll method (for later futures configs): ratio back-adjusted continuous series with a fixed roll rule (roll 5 trading days before expiry by default), or Databento's continuous-contract series when that data is used. Method and roll dates are written in `docs/BACKTESTING.md` and in every snapshot manifest. Yahoo `=F` series are display-only and never used for backtests.

Paper sim (v1.0.5): same fill model on delayed bars, labeled "simulated fills" everywhere. No live money ever routes from this app.

Regime filter (v1.0.8, tested as a hypothesis): a daily 3-state regime label (bearish, range-bound, bullish) for SPY and GLD (BTC once Crypto ships in v1.1) from a Hidden Markov Model (`hmmlearn`), stored in `market_context` and usable as a strategy filter. The model is refitted inside each walk-forward window on past data only. It stays only if it improves a strategy after costs.

Historical data plan

Price, macro and scheduled-event strategies can be backtested from day one; news-sentiment and surprise-tag strategies wait until our own recorded history is long enough.

Data	Source	Depth	How
Daily bars: stocks, ETFs, indices, VIX (Yahoo futures for display only)	Yahoo (yfinance), Tiingo as check	10–20 years	One backfill script → Parquet snapshot + hash
Intraday bars	Yahoo (recent weeks only) + own recorder	Grows from week 3	Daily job appends to `bars_intraday`
Intraday history (optional)	Databento, 1-minute OHLCV, ES + SPY first; more symbols only if the cost estimate allows	As many years as the credits cover	One-time purchase within the $125 credits, cost estimate checked first
Rates, CPI, yields, broad dollar index (DTWEXBGS, from 2006; ICE DXY is not on FRED)	FRED, with ALFRED vintages	Decades	Values as first published, to avoid look-ahead
Positioning	CFTC COT	Back to 1986	Weekly series in Parquet
Scheduled events	Fed (FOMC dates), BLS (CPI/NFP dates)	10+ years	Backfilled into `economic_events` with scheduled = true
Earnings dates	FMP / Finnhub	Limited on free tiers	Forward-collected; history only where available
News + tags	Own ingestion	From week 3	Stored with both `published_at` and `ingested_at`
Surprise geopolitical tags (optional)	GDELT	Back to 2015	Rough backfill only if a strategy needs it; adds maintenance

Rules

- Snapshots are frozen. Yahoo revises data; a backtest always reads a named snapshot, never live data.
- Point-in-time: news uses `ingested_at` for backtests, and macro uses first-published values.

- Splits and dividends: store raw OHLC plus adjusted close as separate columns, method written in the manifest. Futures: roll method and roll dates written in the manifest.
- A small frozen sample (the golden day + a few weeks of bars) is committed to `sample-data/` for CI.

Evaluation and testing

Three layers are checked separately: deterministic code in CI on every push, Hermes with labeled evals run locally or weekly, and signals with statistical checks; results go into the README.

1. Code tests (pytest, CI on every push)

- Unit tests: importance score, dedupe, tagger rules, indicators, composite score, fill model, fees/spread, sensitivity joins, ticker ID mapping, UTC/session handling.
- Golden-day test: one recorded day in `sample-data/` must always give the same ranking and tags.
- Determinism test: a backtest run twice on the committed sample gives byte-identical metrics. Multi-timeframe test: higher-timeframe structure uses only candles already closed at signal time.
- Schema tests: malformed or partial Hermes JSON is caught and never stored.
- Failure injection: a source goes down → real error, alert, stale label, zero fabricated rows.
- Source contract tests: recorded API responses checked against expected shape, so a provider change shows up as a failing test.

2. Hermes evals (scripts in `backend/scripts/`, results in `backend/eval/`)

Eval	Dataset	Metric	Starting target
`eval_condenser.py`	100 hand-labeled news items	Accuracy per field: sentiment direction, catalyst type, tickers	Direction ≥ 75%, tickers ≥ 90%
`eval_models.py`	Same set	Accuracy, latency, cost per model (Ollama vs OpenRouter)	Picks the cheap tier
`eval_tagger.py`	100 labeled items	Precision + recall per driver tag; direction accuracy	Precision ≥ 70% per tag

Eval	Dataset	Metric	Starting target
`eval_numbers.py`	Narrator + Fed outputs on fixed inputs	Share of outputs with a number not in the input	0%
`eval_ask.py` (v1.0.5)	Fixed synthetic database	Tool choice, correct answer, refusal	Set at first run

Targets are starting points, revised once the first baseline exists. No LLM-as-judge.

3. Signal checks

- Composite sentiment score vs next-day and 5-day returns (rank correlation), refreshed monthly once 60+ days are recorded.
- Importance ranking vs one hand-ranked day per month.
- Backtester cross-check against backtesting.py.
- One `RESULTS.md` per strategy in `backend/eval/strategies/`, including failed ones.

Regression gate: `check_regression.py` compares each eval with a committed baseline. Locally it uses real numbers; in CI it uses a deterministic mock, so CI never calls an LLM.

Data privacy: real news text and labels from paid or copyrighted sources stay git-ignored; the repo commits label files with IDs and a small synthetic set.

CI (GitHub Actions, `ci.yml`): ruff lint → migrations on a PostgreSQL service container → pytest (unit, golden day, determinism, multi-timeframe, schema, failure injection, source contracts) → regression gate with mock → frontend build. Weekly scheduled workflow (optional): source contract checks.

Repo structure and docs

Same skeleton as Receipt Ghostbuster, plus folders for strategies, data snapshots and evals.

```
backend/
  app/
    ingest/      one adapter per source, behind a common interface
    scoring/     dedupe, importance, composite score
    tagger/      driver rules, sensitivity matrix
    hermes/      prompts (versioned), Pydantic schemas, client, number check
    market/      quotes, futures strip, sector heatmap, Fed panel
    backtest/    indicators, fill model, engine, metrics
    data/        DuckDB access, snapshot + manifest tools
    api/         FastAPI routes
```

```
  tests/         unit, golden day, determinism, schema, failure injection,
contracts
  scripts/       backfill.py, eval_condenser.py, eval_models.py,
eval_tagger.py,
                 eval_numbers.py, eval_ask.py, check_regression.py,
load_sample_data.py
  eval/          RESULTS.md, MODELS.md, tagger/, numbers/, ask/, strategies/
frontend/        React + Vite
strategies/      JSON configs (one file per strategy)
db/              PostgreSQL migrations (Alembic) + seeds (taxonomy,
sensitivities, sources, tickers)
n8n/             workflow exports (triggers only) + README
data/            Parquet snapshots (git-ignored)
sample-data/     golden day, small bar sample, synthetic eval sets
docs/            DATA_SOURCES.md, BACKTESTING.md, MONITORING.md, PRIVACY.md,
design/
.github/workflows/ ci.yml, weekly.yml
README.md  APP_OVERVIEW.md  DECISIONS.md  CHANGELOG.md  HANDOFF.md
roadmap.md
```

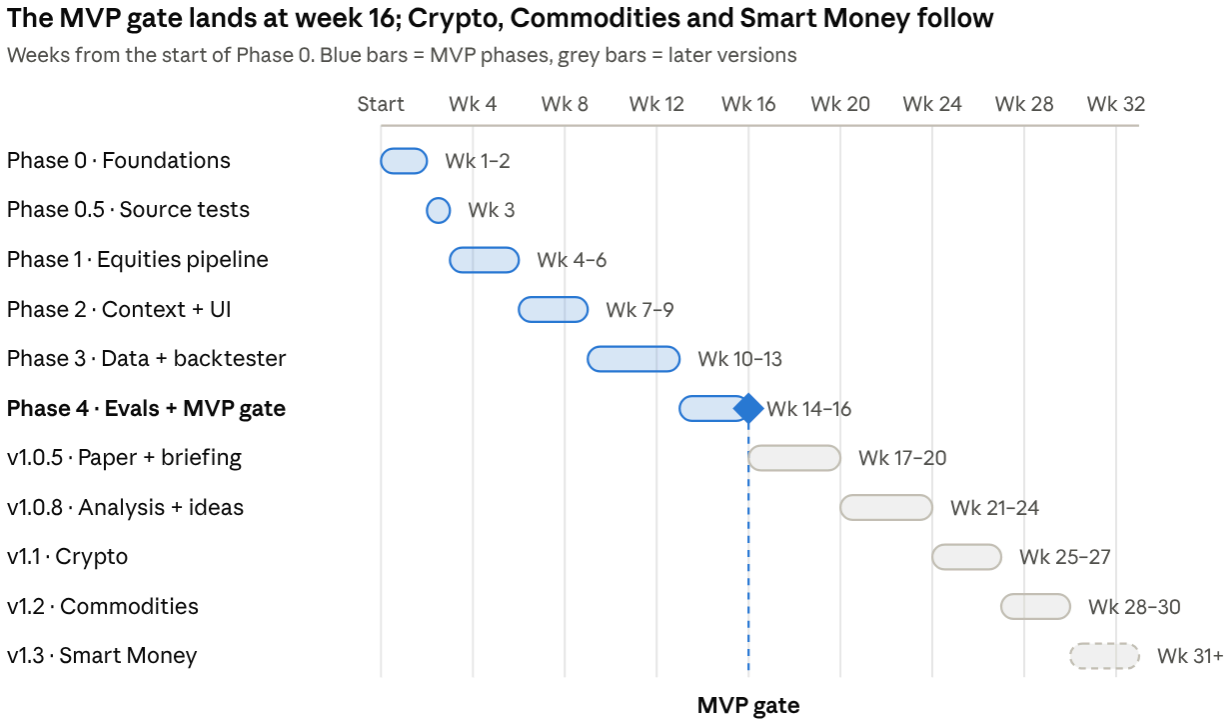
Docs

File	Holds
`README.md`	Problem, features by version, principles, architecture, **Results** (condenser accuracy, tagger precision/recall, invented-number rate, model comparison, backtests with caveats), how to run, demo
`APP_OVERVIEW.md`	What each section does
`DECISIONS.md`	Every locked or rejected decision with the reason
`CHANGELOG.md`	One entry per version
`HANDOFF.md`	Current state, next step, known issues, for picking work back up
`docs/DATA_SOURCES.md`	Each source: limits, terms, fallback, last verified date
`docs/BACKTESTING.md`	Method, fill model, promotion thresholds, known biases, futures roll method, no-USO rule
`docs/MONITORING.md`	`llm_traces`, source health, alerts, budget, backup status

File	Holds
`docs/PRIVACY.md`	What stays local, Sentry scrubbing

Version roadmap

The MVP gate falls at week 16; the five later versions take roughly another 17 weeks at part-time pace.



TradeHub roadmap · 6 MVP phases, 5 later versions

Weeks are estimates for a part-time solo builder. A phase starts only when the previous one passes its acceptance list, so dates move but the order does not.

Implementation phases

Six MVP phases over about 16 weeks, then five follow-on versions; each phase ends with a checkable acceptance list.

Phase 0 — Foundations (weeks 1–2)

- [ ] Repo scaffold with the structure above; README, DECISIONS, HANDOFF, APP_OVERVIEW, CHANGELOG
- [ ] Docker Compose: PostgreSQL 16, FastAPI, SearXNG, n8n (existing), Ollama (existing)

- [ ] PostgreSQL container with a persistent volume, port closed to the outside, Alembic migrations for v1.0 tables, seeds: taxonomy, sensitivities, sources, tickers with canonical IDs; nightly pg_dump backup from day 1
- [ ] Secrets in n8n credentials / env files, never in git
- [ ] CI skeleton: ruff + pytest + frontend build
- [ ] Optional: apply for Reddit API access (non-blocking; nothing in the plan depends on it)
- [ ] Write promotion thresholds into `docs/BACKTESTING.md` (copied from v1.4)

Accept: schema migrates clean; CI green; seeds loaded; compose boots end-to-end.

Phase 0.5 — Source validation + start recording (week 3)

- [ ] For every v1.0 source: working key + one successful call, logged in `docs/DATA_SOURCES.md` with date and limits
- [ ] Verify: StockTwits access, Finnhub's free earnings calendar, each FMP `/stable/` endpoint on a free key, Stooq key, Tiingo key
- [ ] Record the fallback decision for each failure (e.g. Finnhub, then yfinance earnings dates, if FMP lacks the earnings calendar)
- [ ] Start the daily recorder: news headlines + daily and intraday bars for the watchlist
- [ ] Run the first daily backfill and freeze snapshot v1

Accept: every source marked working / fallback / dropped; recorder running daily; snapshot v1 in `datasets`.

Phase 1 — Equities pipeline (weeks 4–6)

- [ ] Ingest adapters: Yahoo RSS, Finnhub news, StockTwits (if working), SearXNG
- [ ] Dedupe (URL + near-text similarity), importance score, driver tagger v1
- [ ] Hermes condenser (cheap tier) + Pydantic + retry-once + `llm_traces`
- [ ] Composite sentiment score per ticker per day
- [ ] Start labeling: 100 news items for condenser and tagger evals
- [ ] Golden day recorded and committed

Accept: 3 consecutive days of hands-off ingest; zero fabricated rows; dedupe rate sane; 100% of Hermes rows valid or logged as failures; golden-day test green.

Phase 2 — Context layer + UI (weeks 7–9)

- [ ] Quotes and futures strip with delayed labels; sector heatmap from sector ETFs

- [ ] Fed & macro panel: FOMC/CPI/NFP calendar, FRED prints and rate history, CME FedWatch external link
- [ ] Earnings dates on Ticker Pages and watchlist
- [ ] Dashboard, Market Pulse Equities tab, Ticker Pages, Settings with source health
- [ ] Hermes importance narrator (top 5) + Fed explainer, both behind the number check

Accept: TSLA, SPY and GLD pages complete; strip updates on refresh; all delayed data labeled; next 3 macro events visible with countdown; no Hermes text shown that fails the number check.

Phase 3 — Historical data + backtester (weeks 10–13)

- [ ] Full backfill: 10–20 years daily bars, FRED/ALFRED, COT, FOMC/CPI/NFP dates → frozen snapshot
- [ ] Optional: Databento 1-minute backfill within the free credits (ES + SPY first)
- [ ] Strategy-config engine with `structure_tf` / `entry_tf` + building blocks v1
- [ ] Backtester with fill model (next-open and limit-touch fills), metrics, `backtests` rows tied to dataset + config hash
- [ ] Gap fade first: proves the engine; cross-checked in backtesting.py
- [ ] Price action supply/demand, VIX spike, FOMC drift, Gold + Dollar + COT on the tuning period
- [ ] SLC on weekly + daily (alternative)
- [ ] Optional: Sortino, Calmar and max drawdown in results
- [ ] Strategy Lab UI: configs, backtest results, equity curve

Accept: reruns byte-identical (CI); gap fade cross-check matches within tolerance; multi-timeframe look-ahead test green; one `RESULTS.md` per strategy run, failures included.

Phase 4 — Evals, hardening, MVP gate (weeks 14–16)

- [ ] Run condenser, model-comparison, tagger and number evals; commit baselines; decide the cheap tier
- [ ] Regression gate in CI with mock
- [ ] Retry/rate-limit guards per source, failure-injection tests, backups (pg_dump + configs + Parquet snapshots) with a tested restore
- [ ] README Results section filled

MVP gate (v1.0)

1. 5 consecutive weekdays of hands-off pipeline, no babysitting.
2. Dashboard shows market state and what's hot at a glance.

3. 6 starter strategies defined (SLC as the alternative); at least 4 backtested on a frozen snapshot, reproducible, one cross-checked.
4. Eval baselines committed; condenser and tagger at or above starting targets, or the gap written down with a fix plan.
5. Zero invented numbers shown in the UI.
6. LLM cost under €5 for the month (from `llm_traces`).
7. Any failure shows a real error + alert, never fake data.

After the MVP

Version	Weeks	Scope	Accept
v1.0.5 Paper + briefing	17–20	Live paper sim, journal, morning briefing, Ask-Hermes, ticker summary, favorite strategies	2 strategies paper-running with journaled trades; Ask eval baseline; briefing passes number check 5 days in a row
v1.0.8 Analysis + suggestions	21–24	Deep Analysis (FMP fundamentals, computed scores, Hermes report), strategy suggester → hypothesis → backtest → one-tap approve, weekly review, A/B mode, regime label + filter (hypothesis)	3 Deep Analysis reports; one hypothesis taken through the full loop
v1.1 Crypto	25–27	CoinGecko, crypto RSS + subs, Fear & Greed, Binance funding rates, 24/7 rolling ingest, BTC/ETH as instruments	72h rolling ingest stable; BTC page + crypto tab complete
v1.2 Commodities	28–30	COT weekly job, EIA, futures context panel, sensitivity matrix in UI, Gold/Oil/Silver pages	Gold page shows tags + drivers + COT + DXY together; COT Friday job runs
v1.3 Smart Money	31+	EDGAR Form 4 / 13F ingest first; Polymarket wallet scan only if reachable from Portugal; optional on-chain extras	Insider filings visible within 24h (whale trades too if Polymarket is reachable)

Ongoing hardening: source health view, backups, docs updated each release, monthly cost review, quarterly re-check of every free tier.

Risks and mitigations

The biggest risks are free sources changing terms and the scope outgrowing part-time hours; both have a planned response.

Risk	Mitigation
StockTwits and Reddit both unavailable	Composite score runs on news sentiment, news volume and tags; social is additive
Polymarket blocked in Portugal	Not used in v1.0: Fed panel is FRED-only; Polymarket re-checked at v1.3
FMP free tier lacks needed endpoints	Earnings from Finnhub, then yfinance; sectors computed from ETFs; fundamentals deferred to v1.0.8
Yahoo unofficial endpoints change	Tiingo for daily, Stooq/Twelve Data for futures; contract tests catch shape changes
A free tier changes or closes	Sources are config; `DATA_SOURCES.md` re-checked quarterly
Hermes invents numbers	Number check blocks the output; tracked in `eval_numbers.py`
LLM cost creep	`llm_traces` budget alert, daily condenser cap, cheap tier for volume
Overfitting strategies	Held-out period, walk-forward, min 30 trades (under 100 = low confidence), costs always on, win rate never decides
Look-ahead bias	Frozen snapshots, `ingested_at` for news, ALFRED vintages for macro
Home-server disk fails or data is lost	Nightly pg_dump + weekly Parquet copy to a second disk or cloud folder; monthly restore test; raw article text pruned after 30 days
Solo burnout / scope creep	MVP gate at week 16; no new section until the current phase is finished
Home server offline	No positions at risk; gap labeled; next run backfills

Monthly cost

Expected running cost is about €2–7 a month, almost all of it hosted LLM calls.

Item	Cost per month
PostgreSQL in Docker (self-hosted)	€0
n8n, SearXNG, Ollama, DuckDB (self-hosted)	€0
Data APIs on free tiers (Yahoo, Tiingo, Stooq, Twelve Data, Finnhub, FMP, FRED, BLS, CFTC, EIA, CoinGecko, Binance, StockTwits, Reddit, Polymarket, EDGAR)	€0
Databento backfill (one-time, within free credits)	€0
OpenRouter Hermes (better tier)	~€2–5
Firecrawl credits (rare, last resort)	~€0–2
Total	**~€2–7**

Decisions log and open questions

The items below are settled and not to be re-litigated; six open questions are answered by tests, evals or the owner's chart settings, not by opinion.

Rejected

Rejected	Reason
Alpaca / broker execution	Owner trades by hand; removes the riskiest part for a solo dev
Finviz Elite	No free API; screener covered by FMP; revisit only for a low-float strategy
CryptoPanic	Free developer plan ended April 2026
X/Twitter API	Paid, out of budget
StockTitan	No public developer API
Brave Search API	Free tier replaced by metered billing in Feb 2026
Alpha Vantage	25 requests/day is too few

Rejected	Reason
Massive (ex-Polygon), US Treasury API	Overlap with Tiingo and FRED
4th category "Indices"	Covered by the futures strip and index tickers
Per-category LLM prompts	One engine, one template set, different configs
Real-time CME / paid feeds	Delayed and daily data fit this use
Taxonomy beyond ~10 tags	Classification accuracy and dead-weight rule
Supabase	Free plan allows 2 active projects, both already in use; self-hosted PostgreSQL in Docker has no project, storage or pause limits. Price history stays in Parquet + DuckDB
8-week MVP	Too much for part-time; replaced by a smaller MVP at week 16

Open questions

1. Cheap tier: Ollama or a small OpenRouter model — decided by `eval_models.py` in Phase 4.
2. Social sentiment: StockTwits access tested in Phase 0.5; Reddit added only if approval ever arrives.
3. Databento: whether the intraday backfill is worth the credits — decided at the start of Phase 3 from the cost estimate.
4. Crypto paper trading (24/7 sim) in v1.1 or later — decided at the v1.1 start.
5. SLC Stochastic: defaults (14, 3, 3; 80/20; %K) until the owner's chart settings replace them; the 4h + 5m test waits for intraday data.
6. Price-action defaults (markets, first-touch zones, stop-first, shorts allowed) — confirmed or changed after the first Phase 3 run.

Deferred extras (only after v1.2): market breadth card, global sessions on the strip (Nikkei, DAX), GDELT backfill, unusual options flow (no free source).